

SANDESH SHARMA

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[\[LinkedIn\]](#) | [\[Academic website\]](#)

RESEARCH INTERESTS

- **Edge AI & Mobile Computing:** Efficient Neural Network Execution, On-Device Inference & Fine-Tuning, Resource-Constrained Machine Learning (TinyML).
- **Computer Vision:** Real-Time Object Detection, Lightweight Optical Character Recognition (OCR), Autonomous Perception.
- **Efficient Deep Learning Systems:** Model Quantization, Pruning, Parallel Multi-Model Execution on Mobile Hardware (CPU/GPU/NPU).

EDUCATION

Pokhara University

2019 – 2025

Bachelor of Science in Software Engineering

Pokhara, Nepal

- **CGPA:** 3.84 / 4.00
- **Class Rank:** 1st out of 55 students
- **Honors:** Dean's Honor List (Awarded for top academic achievement)
- *Note: Program duration extended by one year due to academic calendar disruptions during COVID-19.*

PROFESSIONAL EXPERIENCE

Exotrac LLC

Oct 2022 – Present

Mobile Applications Engineer

New York, USA (Remote)

- **On-Device AI & Computer Vision Integration:** Deployed and optimized machine learning models on iOS and Android platforms using **TensorFlow Lite (TFLite)** and **ONNX Runtime** for automated yard management systems.
- **Parallel Edge Inference Engine:** Integrated multiple computer vision models to run concurrently on mobile devices under strict latency, memory, and battery constraints—enabling real-time container/trailer detection, code recognition, and OCR.
- **Collision Avoidance & Distance Estimation:** Implemented real-time computer vision algorithms to assist drivers with collision detection and back-up distance calculations using on-device camera streams.
- **Cross-Functional ML Collaboration:** Collaborated closely with the ML research team to evaluate model artifacts, optimize graph execution, and adapt model architectures for low-latency edge deployment.

Codekavya Technologies Pvt. Ltd.

Sep 2020 – Oct 2022

Co-Founder & Mobile Application Developer

Pokhara, Nepal

- Architected and delivered cross-platform mobile applications across diverse domains using Flutter, Kotlin, and Swift.
- Managed client requirements, technical architecture design, software execution, and deployment lifecycles.

ACADEMIC & RESEARCH PROJECTS

CricPulse: Real-Time High-Throughput Scoring & Analytics Platform

2023 – 2024

B.Sc. Major Capstone Project | Supervisor: Dr. Udaya Raj Dhungana

- Designed and evaluated an event-driven, real-time analytics system architecture to handle low-latency score updates and analytics streaming across mobile (Flutter) and web platforms.
- Integrated WebSockets and Supabase/PostgreSQL pipelines to handle real-time data broadcasting, edge runtime execution (Deno), and state synchronization across multiple concurrent client devices.
- Conducted system feasibility analysis, architectural design (UML, sequence flows), manual/automated functional unit tests, and performance benchmarking under high query loads.

Herchaha: Interactive AI-Powered NLU Healthcare Framework

2023

B.Sc. Minor Project | School of Engineering, Pokhara University

- Performed an extensive literature review across 20+ peer-reviewed papers on conversational AI, natural language understanding (NLU), and biomedical data systems.
- Developed an AI chatbot architecture leveraging the RASA framework, utilizing intent classification and entity extraction models for dynamic health queries and self-assessment routines.
- Built RESTful APIs using Django and ReactJS to connect conversational models with real-time public health data streams, hospital appointment databases, and interactive D3.js visualization modules.

TECHNICAL SKILLS

- **AI / Edge ML Frameworks:** TensorFlow, TensorFlow Lite, ONNX Runtime, Roboflow, OpenCV
- **Programming Languages:** Python, Kotlin, Swift, Java, C/C++, Dart, JavaScript
- **Mobile & Web Frameworks:** Flutter, Android SDK, iOS Development, Django, ReactJS, Node.js
- **Database & Cloud Tools:** PostgreSQL, Supabase, Git, REST APIs, WebSockets

HONORS & AWARDS

- **Dean’s Honor List:** Awarded by Pokhara University for achieving top academic standing.
- **Academic Merit Rank 1:** Ranked **1st out of 55 students** in the B.Sc. Software Engineering cohort.
- **All-Nepal SLC Rank 2:** Achieved nationwide 2nd rank in the national high school examination (3.95 GPA).
- **Select Distinguished Toastmasters Club Award:** Recognized during presidency for outstanding club growth and performance (2025–2026).

LEADERSHIP & EXTRA-CURRICULAR ACTIVITIES

Pokhara Toastmasters Club

2025 – 2026

Club President

Pokhara, Nepal

- Led organizational recovery initiatives, successfully doubling active club membership and attaining the **Select Distinguished Club** status.
- Mentored 15+ members in public speaking and executive communication; completed **Toastmasters Pathways Level 4**.
- *Previous Roles:* VP of Education (2024–2025), Secretary (2023–2024)—managed educational program schedules, meeting governance, and documentation.

Pokhara University Tech Club

2020 – 2022

Active Member

Pokhara, Nepal

- Organized technical workshops, hackathons, and software engineering peer-mentorship sessions.

STANDARDIZED TESTS & LANGUAGES

- **IELTS Academic:** Overall Band **8.0** (Listening: 8.5, Reading: 8.5, Writing: 7.0, Speaking: 7.5)
- **Languages:** English (Fluent), Nepali (Native), Hindi (Fluent)